Deepfake Audio Detection Using Machine Learning and Audio Signal Processing

Abstract

The rapid advancement of generative artificial intelligence has enabled the creation of highly realistic synthetic speech, commonly referred to as deepfake audio. Such audio can be exploited for impersonation, fraud, misinformation, and social engineering attacks. This project presents a machine learning based system for detecting AI-generated speech and distinguishing it from genuine human speech recordings. The proposed approach utilizes advanced audio feature extraction techniques and a Random Forest classifier to identify artifacts present in synthetic audio. Experimental results demonstrate high classification performance with an accuracy of 97.47%, F1-score of 97.50%, and Equal Error Rate (EER) of 2.93%.

1. Introduction

Deepfake audio generation has become increasingly sophisticated due to advances in deep learning and text-to-speech technologies. Modern synthetic speech systems can produce highly realistic voices that are often difficult for humans to distinguish from genuine speech. Consequently, robust deepfake audio detection systems are essential for cybersecurity, digital forensics, media verification, and fraud prevention.

The objective of this project is to develop a machine learning based framework capable of classifying audio recordings into two categories:

Genuine (Human Speech)

Deepfake (AI-Generated Speech)

2. Dataset Description

The project utilizes the Fake-or-Real (FoR) Dataset, specifically the for-norm version. The dataset contains both genuine human speech recordings and AI-generated speech samples collected from multiple speech synthesis systems.

The dataset is balanced and normalized in terms of sample rate, volume, and channel configuration, making it suitable for machine learning based classification tasks.

Training and evaluation were performed using randomly sampled genuine and deepfake audio recordings from the dataset.

3. Audio Preprocessing

The following preprocessing steps were applied:

Audio files were resampled to 16 kHz.

Audio duration was standardized to 2 seconds.

Short audio samples were padded with zeros.

Long audio samples were truncated to maintain a fixed input length.

These preprocessing operations ensured consistency across all training and testing samples.

4. Feature Extraction

A comprehensive feature extraction pipeline was implemented using the Librosa library.

The extracted features include:

MFCC (Mel Frequency Cepstral Coefficients)

Delta MFCC

Chroma Features

Spectral Contrast

Tonnetz Features

Spectral Centroid

Spectral Bandwidth

Spectral Rolloff

Zero Crossing Rate

Root Mean Square (RMS) Energy

These features capture important spectral, temporal, harmonic, and energy-related characteristics of speech signals and provide strong discriminative information for distinguishing genuine and synthetic audio.

5. Machine Learning Model

The extracted feature vectors were normalized using StandardScaler and subsequently used to train a Random Forest classifier.

Model Configuration:

Classifier: Random Forest

Number of Trees: 600

Maximum Depth: 30

Minimum Samples Split: 4

Minimum Samples Leaf: 2

Random State: 42

The Random Forest model was selected because of its robustness, ability to handle high-dimensional feature spaces, and strong generalization performance.

6. Evaluation Metrics

The model was evaluated using the following performance metrics:

Accuracy

Precision

Recall

F1 Score

Confusion Matrix

Equal Error Rate (EER)

These metrics provide a comprehensive assessment of classification performance and detection reliability.

7. Experimental Results

Accuracy: 97.47%

F1 Score: 97.50%

Equal Error Rate (EER): 2.93%

Confusion Matrix:

721 29

9 741

Class-wise Performance:

Genuine Speech

Precision: 0.99

Recall: 0.96

F1 Score: 0.97

Deepfake Speech

Precision: 0.96

Recall: 0.99

F1 Score: 0.97

The obtained results significantly exceed the required benchmark criteria and demonstrate strong detection capability for both genuine and AI-generated speech samples.

8. Applications

Deepfake Audio Detection

Digital Media Verification

Audio Forensics

Voice Authentication Systems

Cybersecurity Applications

Fraud Prevention Systems

Content Verification Platforms

9. Conclusion

This project successfully developed a high-performance deepfake audio detection system using machine learning and advanced audio feature extraction techniques. By leveraging spectral, harmonic, and temporal speech characteristics, the Random Forest classifier achieved an accuracy of 97.47%, an F1-score of 97.50%, and an Equal Error Rate of 2.93%.

The experimental results demonstrate that feature-based machine learning approaches remain highly effective for deepfake audio detection and can serve as practical solutions for real-world security and verification applications.

10. Future Work

Integration with the ASVspoof 2019 dataset.

Cross-dataset evaluation and benchmarking.

Ensemble learning approaches.

Real-time inference optimization.

Transformer-based audio classification models.

Deployment on cloud-based audio verification platforms.

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```
acc = accuracy_score(
    y_test,
    preds
)

f1 = f1_score(
    y_test,
    preds
)

print("Accuracy =",acc)

print("F1 =",f1)
```

```python
cm = confusion_matrix(
    y_test,
    preds
)

print(cm)
```

```
[[721  29]
 [  9 741]]
```

```python
fpr, tpr, _ = roc_curve(
    y_test,
    probs
)

fnr = 1 - tpr

eer_idx = np.nanargmin(
    np.abs(
        fpr - fnr
    )
)

eer = (
    fpr[eer_idx]
    +
    fnr[eer_idx]
) / 2

print(
    "EER =",
    eer
)
```

```
EER = 0.029333333333333333
```